THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS · HARD LEVEL MCQ TEST SET 2

Full Syllabus (Chapters 1–10) | RBSE Pattern | Session 2026–27

Time: 1 Hour Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–50 are Multiple Choice Questions (1 mark each); there is no negative marking.
3. Choose the correct option and write its letter (a, b, c or d) against the question number.
4. Questions cover the whole syllabus of the class.
5. Hard level: application and higher-order questions.

MULTIPLE CHOICE QUESTIONS — 50 × 1 = 50

Choose the correct option and write it in your answer book.

1. A pattern adds 1, then 2, then 3, ... Starting at 10, the term after 12 additions is: 1

(a) 22 **(b)** 76 **(c)** 154 **(d)** 88

2. 4 lines are drawn through a point O. The number of straight angles at O (pairs of opposite rays) is: 1

(a) 5 **(b)** 4 **(c)** 12 **(d)** 3

3. The number of digits in the largest palindrome smaller than 10000 is: 1

(a) 3 **(b)** 4 **(c)** 6 **(d)** 5

4. A bar graph shows sales of ₹2000, ₹3500, ₹2500. If the scale is 1 cm = ₹500, the total height of all three bars is (cm): 1

(a) 14 **(b)** 16 **(c)** 8000 **(d)** 18

5. Which is the greatest 3-digit number that leaves remainder 3 when divided by 7? 1

(a) 990 **(b)** 997 **(c)** 1004 **(d)** 998

6. Tiling a rectangular floor of length 9 m at ₹150 per m^2 costs ₹6750. Its breadth (m) is: 1

(a) 10 **(b)** 14 **(c)** 5 **(d)** 45

7. Which is the smallest of $\frac{5}{7}$, $\frac{7}{10}$, $\frac{3}{4}$, $\frac{8}{11}$? 1

(a) $\frac{5}{7}$ **(b)** $\frac{3}{4}$ **(c)** $\frac{7}{10}$ **(d)** $\frac{8}{11}$

8. Three circles of radius 3 cm are drawn in a triangle arrangement, each touching the other two. The triangle joining the centres has perimeter (cm): 1

(a) 27 **(b)** 18 **(c)** 15 **(d)** 21

9. A regular polygon has 4 sides. The number of different angles (less than 360°, other than 0) through which it can be turned to coincide with itself is: 1

(a) 3 **(b)** 5 **(c)** 2 **(d)** 4

10. The value of (-17) + 17 + (-6) is: 1

(a) -5 **(b)** -7 **(c)** -6 **(d)** 11

11. The sum $1^3 + 2^3 + \cdots + 9^3$ equals: 1

(a) 2025 **(b)** 90 **(c)** 45 **(d)** 729

12. Rays OA, OB, OC, OD make equal angles around O (a full turn). The measure of each angle is: 1

(a) 82 **(b)** 162 **(c)** 72 **(d)** 62

13. On reversing and adding 187, the first sum is: 1

(a) 957 **(b)** 1067 **(c)** 979 **(d)** 968

14. A pictograph has symbols worth 12 each. To show 100, how many FULL symbols are needed (plus one part-symbol)? 1

(a) 10 **(b)** 7 **(c)** 9 **(d)** 8

15. The sum of the first 3 prime numbers after 10 is: 1

(a) 41 **(b)** 43 **(c)** 40 **(d)** 42

16. Out of all rectangles with perimeter 32 cm having whole-number sides, the greatest area (cm^2) is: 1

(a) 72 **(b)** 64 **(c)** 56 **(d)** 80

17. A rod $3\frac{1}{2}$ m long has $\frac{3}{4}$ m cut off. The remaining length in metres is: 1

(a) $2\frac{1}{4}$ **(b)** $4\frac{1}{4}$ **(c)** $2\frac{3}{4}$ **(d)** $\frac{1}{4}$

18. An isosceles triangle OAB with OA = OB has $\angle AOB = 40^\circ$. Each base angle (degrees) is: 1

(a) 75 **(b)** 65 **(c)** 70 **(d)** 80

19. A regular polygon is rotated again and again through its smallest angle 40°. The number of turns needed to complete a full turn is: 1

(a) 11 **(b)** 10 **(c)** 9 **(d)** 8

20. If ? + (-7) = -12, then the missing number ? is: 1

(a) -7 **(b)** -3 **(c)** -5 **(d)** 5

21. The 41st triangular number minus the 40th triangular number is: 1

(a) 41 **(b)** 42 **(c)** 430 **(d)** 40

22. A wheel has 12 equal spokes, so the angle between two neighbouring spokes is 30°. The angle between spoke 1 and spoke 6 (going one way round) is: 1

(a) 120 **(b)** 150 **(c)** 210 **(d)** 180

23. 68 is added to its reverse and the sum is 154. How many reverse-and-add steps in all are needed to reach a palindrome? 1

(a) 5 **(b)** 2 **(c)** 3 **(d)** 4

24. In a bar graph with scale 1 cm = 8 units, the value 60 is shown by a bar of height (cm): 1

(a) 8 **(b)** 7.5 **(c)** 52 **(d)** 7

25. The largest prime number less than 100 is: 1

(a) 96 **(b)** 97 **(c)** 99 **(d)** 98

26. The sides of a triangle are 13 cm, 14 cm and 15 cm. Its perimeter (cm) is: 1

(a) 42 **(b)** 43 **(c)** 55 **(d)** 41

27. To add $\frac{1}{6}$ and $\frac{1}{8}$, the common denominator used is the LCM of 6 and 8. The LCM is: 1

(a) 48 **(b)** 12 **(c)** 30 **(d)** 24

28. The diameter of the circle that exactly fits inside a square of side 10 cm is (cm): 1

(a) 10 **(b)** 9 **(c)** 15 **(d)** 11

29. A regular 6-gon has side 7 cm. Its perimeter (cm) is: 1

(a) 48 **(b)** 49 **(c)** 35 **(d)** 42

30. The value of 7 - 12 - (-7) is: 1

(a) 2 **(b)** 4 **(c)** 12 **(d)** 0

31. At a party, 18 people each hug every other person exactly once. Number of hugs: 1

(a) 171 **(b)** 306 **(c)** 162 **(d)** 153

32. A clock shows 4 o'clock. After 5 hours, the smaller angle between the clock hands is: 1

(a) 90 **(b)** 120 **(c)** 270 **(d)** 150

33. How many steps does the Kaprekar routine take to reach 6174 from 5273? 1

(a) 9 **(b)** 6 **(c)** 7 **(d)** 8

MASTERMIND

34. The total of the table, after removing the largest value, is: 1

Item	Number
Apple	45
Grapes	25
Banana	85
Mango	60
Orange	30

(a) 220 **(b)** 165 **(c)** 245 **(d)** 160

35. The product of the two smallest prime factors of 330 is: 1

(a) 6 **(b)** 5 **(c)** 10 **(d)** 15

36. A square park of side 20 m has a footpath 1 m wide all around outside it. The area of the footpath (m^2) is: 1

(a) 80 **(b)** 84 **(c)** 88 **(d)** 92

37. $\frac{7}{9} + \frac{2}{3} - \frac{5}{18} =$ 1

(a) $\frac{4}{9}$ **(b)** $1\frac{1}{6}$ **(c)** $1\frac{13}{18}$ **(d)** $\frac{7}{18}$

38. Two concentric circles have radii 5 cm and 10 cm. A point is 0 cm from the common centre. It lies in the ring between the circles only if its distance is more than 5 cm and less than 10 cm. How many of the distances 5, 7, 10, 2, 9, 11 cm give a point in the ring? 1

(a) 2 **(b)** 6 **(c)** 4 **(d)** 3

39. The image of a point 5 units from a mirror line is how far from the original point (units)? 1

(a) 5 **(b)** 10 **(c)** 15 **(d)** 11

40. The sum of the integers from -6 to 7 is: 1

(a) 13 **(b)** 7 **(c)** 6 **(d)** 8

41. **Assertion (A):** The sum of the first 5 odd numbers is 25. 1
Reason (R): 25 is a square number.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

42. **Assertion (A):** A straight angle measures 180°. 1
Reason (R): A right angle measures 90°.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

43. **Assertion (A):** In the hailstone (Collatz) sequence, the number after 7 is 22. 1
Reason (R): If the number is odd, multiply by 3 and add 1; if it is even, divide by 2.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

44. **Assertion (A):** In a bar graph all the bars are drawn with the same width. 1
Reason (R): The height (or length) of a bar shows the quantity.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

45. **Assertion (A):** 2 is the only even prime number. 1
Reason (R): Every even number greater than 2 has 2 as a factor other than 1 and itself.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

46. **Assertion (A):** The area of a square of side 7 cm is 49 cm^2. 1
Reason (R): Area is measured in square units.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

47. **Assertion (A):** The fraction $\frac{7}{4}$ is an improper fraction. 1
Reason (R): $\frac{7}{4} = 1\frac{3}{4}$.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

48. **Assertion (A):** A circle has diameter 14 cm. Its radius is 7 cm. 1
Reason (R): The radius of a circle is twice its diameter.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

49. **Assertion (A):** A rectangle that is not a square has 2 lines of symmetry. 1
Reason (R): Such a rectangle has rotational symmetry of order 2.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

50. **Assertion (A):** The sum of 5 and -5 is 0. 1
Reason (R): Zero is neither positive nor negative.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

END OF QUESTION PAPER **Total: 50 Marks**